

ActInf MorphStream 002.1 ~ Alon Loeffler "From Neuromorphic Nanowire Networks to Active Inference"

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Okay. Hello, welcome everybody to the Active Inference Institute's second episode of our Morph Stream series. So these sessions are intended as a space to showcase and discuss the intersection between neuromorphic computing, so brain-inspired computing, and active inference. If you're new to neuromorphic computing, I did give a short primer in the first session, which is available on YouTube. So the format is that our guest speaker will present in the first part, then there's a Q&A, and then a wider conversation for the rest of the call. So our first ones featured David Cappell, who's kindly joined us again today. Thank you, David. And he gave an absolutely fascinating talk about a free energy model to uncover the role of noise in the brain. So it's like active inference at the synapse. And today's episode features Alon Loeffler. He's a postdoctoral scientist at Cortical Labs in Melbourne, Australia. Cortical Labs being the company behind the very interesting DishBrain paper, which recently came out. So he specializes in synthetic biological intelligence and has a PhD in neuromorphic nanowire networks from the University of Sydney. He also has almost three, sorry, over three years of experience in designing learning tasks and algorithms for brain-inspired systems. And today he's going to be presenting some really interesting research on neuromorphic nanowire networks. And so if you want to share your screen. Absolutely. Thank you. All right. Can you see this? All good. Yep. All good. All good. Fantastic. So thank you very much for having me. And thanks for the wonderful introduction. Not much more to say about myself about, so that was a perfect summary. Yeah. Today I'm going to talk mostly about my PhD research, which I finished last year from the University of Sydney, which was in neuromorphic nanowire networks. And that's quite a mouthful. We'll definitely get into what that is. Don't worry. Hopefully by the end, you'll be experts. And I'll also touch a bit at the end about what I'm working on at Cortical Labs at the moment. And there's quite an interesting link. They're slightly different, but hopefully we'll get through that. So just a bit of an overview of how today is going to go. I'm going to give a very brief background. I imagine a lot of the people who are interested in this kind of topics know quite a lot about AI and the state of AI at the moment. But just in case you don't, I'll give some background and go over that. Explain what nanowire networks are. Explain what we actually do, how we model these systems, and how we actually apply these systems. And then at the end, just give a brief overview of what I've been working on for the last year or so. So let's start with a bit of a background. So the brain, the original inspiration for artificial neural networks, AI, kind of this machine learning idea behind AI, was the neuron. The most basic ANN is called a perceptron. And it was sort of in the 50s and 60s. It was created to mimic the idea of this inputs into a neuron from this sort of dendrites into the, through the cell body, into the axon, and then across as an output into different neurons. And that's sort of the idea that we have. That's sort of what became this beginning of artificial neural networks and AI in general. And so, but there's quite a few key differences between neurons and the

structure of neurons in the brain and how we implement our perceptron slash ANN interpretation of these neurons. Mainly, there's the idea behind von Neumann architecture, which is what, how we pretty much implement all of our computing these days, compared to how the brain is structured. So in most modern AI systems, we implement this software of AI on hardware, von Neumann hardware, where we have this separation between computing, processing, and memory. And we have to bus information back and forth from the CPU to the memory, which is quite a big bottleneck. It takes quite a lot of time to do that. And that's pretty much the main kind of limitation of this architecture. Whereas in the brain, there's sort of this integrated idea of memory and processing that happens in the synaptic kind of gaps between neurons, in neural cells, in chemical processes, all that kind of stuff occurs simultaneously. And so we're trying to overcome this bottleneck of this bussing information back and forth, which actually takes quite a lot of energy and quite a lot of time. The main way we've been doing this is adding more transistors. This is sort of how the structures are built using transistors. More of those transistors are added so that we can have more and more computer processing. This is what's called Moore's law. And you may have heard of that before, but the idea is over the past few decades from the beginning of the perceptron, which I introduced, we've sort of seen this two year doubling idea of computer processing power and energy requirement. And then in the last sort of decade or so, we've seen that this is not linear. Well, it's still linear, but it's not quite as simple, this relationship. It's not a doubling of energy and power. We're seeing this 3.5, 3.4 month doubling of requirement for power. So more and more and more transistors are required to do AI and ANN kind of processing and more and more power is therefore required, which is a big issue. Why is this a big issue? Well, firstly, I'll get back to that in a bit of a second, but importantly, another important aspect between the brain and Neumann architecture, another difference is this brain circuitry idea. And so while modern applications of AI are really, really, really good at object recognition and pattern recognition, they're able to beat humans in lots of things like that. But their original neuron-inspired architecture, which I discussed from the perceptron, has transformed into a much more complicated system that's reliant on these higher and higher power requirements and energy requirements, and are held back by this bottleneck. And the feed forward structure that we see that we all quite familiar with, where you have this input layers into hidden layers into an output layer where information flows in from the left, goes through these layers and out to the right. It's an inherent approach in AI, there is some recurrence that can be introduced, but even that's quite a simple method. And this is highly, highly different to biological neural networks. We see a very different structure, we see a very different kind of functionality, even in these recurrent networks. And so the question is, you know, humans behave, at least sometimes, intelligently, with significantly low energy requirements, you know, you and I only need a few meals a day worth of energy, the brain works at around 20 watts of information, of energy, sorry. And in order to try and capture this kind of intelligence and processing capacity of the brain, we need to try and overcome these bottlenecks and structural issues that I've mentioned briefly, that AI faces. And so to do that, we can turn to the brain for inspiration. And then the question that I was asking during my PhD is how can we capture the physical nature of biological intelligence? So that's a

brief background of where the state we're in. And this is where neuromorphics enters, right? That's the background

behind what we were just what the state we're in. But neuromorphics is one suggestion into how to solve this. So neuromorphics, the term neuro comes from neuron, morphic meaning mimicking or copying. Neuromorphics sorted into two main categories. One is the idea of neuromorphic chips, which is something that is probably more familiar to, similar to, you know, computer chips, CMOS, electronic, electrical chips, stuff like that. And they're used as mainly AI accelerators. So how do we implement current AI models, the software on this new hardware, to get more neural mimicking behavior and reduce power

requirements? And something that other people might be more familiar with is spiking neural networks, where the

idea is we take traditional artificial neural networks, and we turn them into this binary input output, or binary inputs and binary outputs, where it's just ones and zeros. And so you get significantly reduced amount of data that you have to

deal with. The weights are significantly more simple to deal with. And so this is a much easier kind of dataset to deal with, and uses a lot more, a lot less energy as well. On the other hand, we have this memristive systems. And so memristive systems are systems that have this thing called conductance-based synapses, where the idea behind it, the word memristor or memristive comes from memristor, which is a variable

resistor. And pretty much it exhibits memory as the kind of current hysteresis in a current response curve.

That might not mean anything, but I'll explain that to you in a little bit. Pretty much the idea is that we get these memristive systems that are much more biological-like when they're self-assembled, much more array-like when the array assembled, but they have this more non-volatile, they have this more interesting behavior than neuromorphic chips do. So there's some examples of these type of systems. Intel and IBM mainly work on AI accelerators in Lohi and True North. Through the Human Brain Project, we have developed

Spinnaker, Brain Scales, and a few other models to try and implement spiking neural networks, which try and replicate spikes in the brain, neurons. And then memristive crossbar arrays are this idea where you have these sort of metal bottom and top electrodes. You have this cell in the middle and you get flow of energy, electricity through that. I'll explain a lot more about this later. It's just sort of a bit of an over-

The thing that we are focusing on today though is something called nano-wire networks, which fall into the self-assembled networks, which means they assemble into a network from the bottom up rather than what you

might imagine these other three categories do in this kind of top-down clean lab fabrication experience. And that makes them a lot more similar to biological systems. So that's a big mouthful, but we'll get, we'll break it down a bit more. So what are nano-wire networks? Well, nano-wire networks are typically made out of a simple material, like a silver material. They're nano, so they're really, really tiny on the orders of nanometers in width, about a thousandth of a hair, width of a hair. And they get to about five to 10 micrometers in length. So they're really long and very, very thin.

So they're really, quite similar to a neuron. And on the right here, you see an example of what a neuro, a nano-wire network looks like, a silver nano-wire network. So made out of silver material, compared to an example of what a biological system looked like. So there's quite similarities, quite significant similarities in the structure of how they look. So what happens is we deposit these nano-wire, these silver wires onto a glass substrate, usually, and they self-assemble. So they assemble into this network and they start to cross over each other. So you can see wherever, let me point, get the pointer out. Where is the pointer?

Sorry. There we go. So wherever a wire crosses over another wire, we assume that there's a junction or a synapse, similar to where a dendrite, an axon connects into a dendrite in neural structures. And so the idea behind this, the structure resembles itself and the function also resembles itself. That's what these two graphs at the bottom look like. So when we zoom into these kind of junctions, as you can see here, that's a zoom in of that. We get this silver, two silver wires are represented here. We get this insulating polymer in between. It's just a simple plastic called PVP polymer in between. And then when that is off, meaning when there's no electric current being passed into the system, nothing special happens. The wires aren't actually touching each other. There's no electrical conductivity. But then when we turn it on, when we feed in some kind of input, voltage input pulse, like is shown here, we see this ion migration, this flow of atoms, positively charged atoms from one of the wires to the other through this plastic polymer. And this is what we call like a switch. It's similar to what an action potential kind of looks like. Where action potentials occur, and then you get this increase in conductivity in the membrane of the neuron, we see a similar behavior in nanowires. This is sort of the basic blocks of how neurons communicate with each other. For those of you who don't know how that happens. And we see a very, very similar kind of response occurring in these nanowires. Now, I mentioned memristors before. Memristors have this property of when this switch-like behavior occurs, when you get this nanofilament forming between across here, you get it's a non-linear activation. So it's not predictable. It's quite a difficult thing to model. And when it forms, it has memory for previous times, previous signals that have occurred that have been input into the system. So it forms at a certain pace, a non-linear pace. Once we stop that pulse, it starts to decay slowly back across the network at a different rate than what it originally formed. So you get these multiple complex behaviors occurring. And because of that decay, you slowly, like the ions decay very slowly. And imagine you've stopped the signal for a couple of minutes, the bridge has decayed halfway, and then you start the signal again, that bridge is going to form a lot quicker than a bridge that hasn't been formed previously. So that's where we get this memory aspect from it. That's a junction level memory. We also get all the beautiful recurrent behaviors of, you know, information isn't just free forward flowing like in artificial neural networks. We get this kind of recurrent activity that information flows in all directions in these networks, which is really, really important for information processing, biological like information processing. Okay. I've talked a lot more than I thought I would. So

let's power through.

So one really-

You can take your time. It's no problem.

Great. One really important, really interesting thing about nanowires is we can model them in a graphical representation where we take the wires and we represent them as nodes, these actual wires. And whenever there's a synapse or a crossover between the junctions, we can represent those as edges. And so that really helps us to look at the structural properties and also the functional connectivity of these networks.

So this is an example of what we've done with a model. You can take a real nanowire network and you can model the density. You can model the length of the wires, which we can measure relatively difficultly, but we can measure them. And then you can construct a simulated version of it, which is definitely not one-to-one. That's one limitation that we have. It's very difficult to model, but exactly the same, but it is experimentally validated, meaning that the activity that we see in the simulated network, we see similar conductance, similar current, similar voltage in the real nanowires.

And what's really lovely and what I've published in my first KHD paper is you can model these nanowires in a graphical representation, as I mentioned before, and then compare them to other structures that occur in the real world. For example, a fully random network over here, a completely lattice-like network where each node is just connected to its nearest neighbor, and then our nanowire network compared to a very simple biological system, which is called the C-Elegans. So what we see is that our biological nanowire network is much more similar, sorry, our bio-inspired neuromorphic nanowire network is much more similar to biological systems like the C-Elegans than it is to random networks or these lattice-like objects. And these lattice sort of like structures is what we see in typically in CMOS chips or computer chips. A bit more about the modeling, I've explained a bit about how this conductivity, this low to high conductance occurs, but we can model that using a mathematical equation. For those of you who like maths, this is the only equation that I will share in this presentation. I don't like having equations on presentations, but it's a pretty cool thing to see. Pretty much we can model when the voltage is higher than what we have a V-set state, which is shown in this graph here. So the voltage is on the x-axis, the current of the y-axis. Once there's a voltage set, once we reach a V-set and once our voltage is higher than that, we have this kind of switch that is on. So we get in a high conductive phase.

And then when the derivative of this lambda function, which models this kind of filament, is at zero, we have when the voltage is between this set state and the reset state that we have, we get this kind of activity where it slowly decreases, the current decreases and flips.

And then when the voltage is lower than reset, we get back to the off state where the current is at zero. So this is what I was mentioning. Memristas have this pinched hysteresis curve. That's what you see here. This is called the pinched hysteresis curve. And this is sort of the idea where we want to see this kind of nonlinear memoristive activity. That's what makes a nanowire, a neuromorphic system, interesting. Don't worry about it if that's too much jargon for you. It's just a bonus for the maths, whoever likes the maths in here. So a bit more of a visual explanation of how this, what the nanowire's

network is actually doing. This is a model of the network we've simulated, where we have this input electrode up here, which is represented by this green star. And then we have a drain or an output electrode bottom right here. We're feeding in a voltage, a standard voltage bias, simple DC voltage bias. And we start to see each one of these wires, as I mentioned, each one of the wires is modeled by a dot here. And each one of the connections between the wires is modeled by an edge. And you start to see this kind of increase in current flow. That's what this is showing, current flow, across the network from the source to the drain. You see this, what we call a winner-takes-all pathway, where a path is formed where these junctions switch on. So all of these junctions are at that on high resistance state and all the others are not yet. And then you start to see like more parallel paths beginning to activate once that original path is formed, which is a really interesting functionality that we can model. We also see this network level kind of increase in G . G is conductance, so it kind of models the, it's the inverse of the resistance. So pretty much we start at a very, very high resistance state, and then we, or low conductance state. And then at some point, and that point is at the path of least resistance, when the drain and the source and the drain are connected, we see this massive increase in conductivity, massive decrease in resistance. And that's that swap between low resistance to a high resistance state. We see that at a network level, as well as an individual junction level, which is really interesting behavior. This is this sort of nonlinear activation function, which might look familiar to some of you that have played with artificial intelligence before. For those of you who haven't, this nonlinear, non-linearity is sort of how we do our calculations and transformations in hidden states. It's, you know, it's this, there's different types of nonlinear activation functions, but there's the reticular and linear activation functions. Anyway, not important. This is sort of what this system does inherently. We don't have to train it to do that. It just, that is the inherent properties of this network. Whereas in artificial neural networks, we sort of have to play around to get what the best model is to fit that. The memory aspect, which I mentioned also happens at a network level, not just a junction level. So if you remember, I mentioned that each filament has memory for what happened previously. The network also has that memory. And what you can see is when we're feeding a bunch of small pulses for a short amount of time, you get this increase in conductance at the network level, what I explained previously, and then it sort of resets back to its original state after a certain amount of time. That's because each one of those junctions that I've mentioned here that is activated takes a while to decay and go back to its normal state. And eventually it all goes back to normal. But if you increase that, the number, the amount of time that you're inputting this voltage and the number of voltage pulses, you can get the network into this kind of steady state where it doesn't reset fully for quite a long time. And so you see this full network kind of activation level, similar to long term memory or long term potentiation compared to short term memory. These are just some really cool things that we've seen in the network. It's much more complex than I have time to really delve into it. But if you have more questions, please feel free to ask. I can answer as many as needed. And so a lot of background, a lot of kind of information dump, but what can we actually

do with these? What is, why are these cool? You know, why are we even interested in these? Well, firstly, we have to understand how we apply these. So as I said, we take these silver nanowires and we drop cast them, meaning we take a little bit, we take from a pre-created mixture of these PVP silver nanowire networks, and we put them in the middle of an electrode array that we've set up. So an electrode array might look like this, where we have nine electrodes on one side that are measuring the activity from electrical activity from the networks and nine on the other side. And you can kind of program any of these to do anything that you want. Nine and nine is quite small compared to a lot of electrodes, but this is just a start that what we did in our system. And so one potential approach to using these nanowire networks is something called reservoir computing. That's a bit closer to artificial neural network kind of implementations. The idea is you have your inputs, like an artificial neural network, you have your inputs, you have your hidden layers in the middle, and then you have your outputs. The main difference in reservoir computing is the reservoir system or the in between layers are untouched. We do not touch them. In traditional AI, the idea is you use something called backpropagation to change your inputs, to get your outputs. You change the weights of each one of these kind of nodes to try and match your outputs with a desired output that you have. And if the desired output is not similar to your current output, you backpropagate, you change the weights of the middle layers, and then you try and make that a bit closer to the output. That's how traditional ANNs work. In reservoir computing, we just let the network kind of do its thing. We don't change the weights. We just let the inherent physics do what it wants. And then we train an app on the outputs. We train a very simple model to try and understand the outputs from the network. And so we can do the same with physical systems like nanowire networks, where we just take out a traditional reservoir system and we put in our nanowire networks. So what does that look like? Well, in a task which I did in, well, a bunch of my collaborators did prior to me and also I did in my PhD, we have this input signal. It's called the nonlinear transformation task. We feed in a sinusoidal input, just a simple sine wave, to the reservoir system, which is our nanowire networks. We let them do their thing. We don't touch them. They have some physics that they do based on all that kind of interesting stuff that I explained before. We get the linear readouts from that network and we train some weight matrix to try and using linear regression, very, very simple kind of method, to try and convert this input signal into a different output signal. And the only way that that can happen is if the reservoir system in the middle has nonlinear behavior, interesting kind of behaviors pretty much, that can transform that input signal inherently, and that we don't need to train a very complex system to do. The other task, for example, is something called the memory capacity task, where you have a random input signal that you're feeding into the reservoir at each time point, and then you delay your signal by however many steps, and then you're trying to reconstruct the time points and steps backwards. These are just a couple of examples. Don't worry if it's a bit too confusing. So what did I do? Another thing that I did, which was really interesting in my second paper, was trying to combine these kind of interesting structures of nanowire networks, which make them a lot more biological with this reservoir computing model. And so over

here,

we have a nanowire network. All of these use the same nanowire network model, but we just changed the structure. So this is an example of a fully connected two layer nanowire network, not nanowire network, sorry, artificial neural network, a fully random network, a nanowire network, and a biological network called the C elegans. And we performed these two tasks, and we showed that the nanowire networks at the certain values outperform, if not perform as well as other types of networks without the need for these. And most interestingly is these bipartite networks, which we use, or these ANN type structures, perform the least effectively when using our models. So it shows that there is an importance in this biological structure as much as there is in the biological function, which I've been talking about previously. So this kind of combination of the two is the most important aspect. And lastly, last thing I want to talk about about the nanowire networks, this is a very speed run introduction into it, was my final paper in my PhD, which was actually trying to take brain inspired learning tasks and implement them in this biological intelligence. So reservoir computing was one way to try and implement some kind of AI-like task into this, but that's not the point of this talk. And the point of my project was to try and get a lot more brain inspired learning, a lot more brain inspired intelligence, those kind of aspects. And so that brings us to this type of task where I tried to implement a brain inspired learning task, inspired by a psychological task called the NBAC task, where you give a target pattern to participants, then you give them a bunch of different patterns, and then you ask them after a certain number of steps, do you remember the original pattern that I presented to you? And that's what we did to this network, we presented a specific pattern, for example, specific pattern number A, pattern A, and then we gave it a bunch of different patterns. And then we asked it, we wanted to see how many steps back it can remember that. That original step, that original pattern. That's pretty much. And if it recognized the pattern successfully, we gave it some external feedback. So we either punished it or rewarded it. And if it, that was in the, what we call the physical reinforcement category. And if we had no reinforcement, then we didn't punish it at all. We didn't give it any feedback and we just saw what it would do. And so the results, so this is an example of what it looks like a bit more clearly.

So if our target pattern here is pattern A, we give a bunch of patterns, then we give pattern A, then we give a couple of other patterns, and then we ask for pattern A again. So this would be three steps back where N equals three. So the N back would be three. And this would be kind of what we feed into the network. On the left is where the inputs are, on the right is sort of what the output that we wanted to get associated with. And each one of these patterns has a different output that is associated with it. And what we saw is that the accuracy, this is a bit of a difficult thing to look at, but pretty much the square dots are what we saw experimentally, the circle dots are what we saw in simulation, and the reinforcement. So what we see with the reinforcements are giving them reward and punishment is at the top here. And what we see without giving them the reward and punishment is at the bottom. So interestingly enough, this is the number of steps back where we presented that original pattern. So for example, at N back equals seven, the original pattern will be right at the

start, and then it'd be a bunch of other patterns, and then we'd ask it again. And so what we see is that without any reinforcement, there is this gradual decay of memory over time for that pattern. And that makes a lot of sense. Humans are pretty bad at this task as well. We forget quite easily after about seven steps as well. So that's quite an interesting thing. But what's really interesting is when we implement this reward and punishment application, which is based in biological reward and punishment, something called supervised learning and reinforcement learning, these two things combined, we see a massive increase in memory for this pattern and this, this, this, yeah, the target. Importantly, we also saw that activity when we had two targets. So if we had like a target A, and then we swapped and asked for target B,

it managed to remember both of those at the same time, which is really interesting. We haven't pushed that to the

limits, but that's a really interesting aspect. And so that shows that we can get this kind of biologically inspired learning mechanisms to give us some interesting activities in this quite complex biological inspired system. How does that look? I've got a video hopefully that will work on me.

Yep, yep, yep, we got there. Come on, you can do it. Okay, maybe I need to select the sorry pointer options turn off laser pointer play. So this is an example of without the reinforcement. We have our inputs on the right here, which is a bit confusing. And then we have our target pattern in this circle type pattern. So what you wanted to what you'd want to see is a clear path from anywhere around here to this target, meaning that the net the network is learning to to prioritize that pathway over time. Without reinforcement, we're seeing that yes, sometimes that path is being formed, most likely when you know that signal is being implemented, but a lot of the time it's not, it's not really prioritizing. And on the other hand, in the reinforcement paradigm, you'll see pretty quickly, we start off with this sort of random kind of path activity. And then you'll see that over time, this path is remembered by the network and prioritized, you do still get other pathways, but they're slowly kind of weakened over time. But this path is almost always staying there.

And that's the idea behind this reinforcement learning. And so the network has remembered that whenever we're given that input signal, we want to go to that output time, which is really cool to see. I think this is a really cool video. And there's a summary image of what happens over time.

This is flipped. So the inputs are now on the left. Sorry, I know it's annoying. But pretty much the red regions are the regions that are strengthened between this when we give reinforcement and we don't give reinforcement, and the blue regions are weakened. So the network is effectively learning to weaken these pathways and strengthen these pathways, similar to what you might, you know, you might say it's similar to this pruning activity or strengthening activity in a synaptic kind of brain like activity and strengthening of synapses, which is really, really interesting, really cool that we managed to see this.

So that's quite a lot of a lot of information. How does this even link into what I'm doing now? So that was my PhD. Right now, I'm working on real neurons at cortical labs. And, you know, neuromorphics are really cool, especially neuromorphic nanowire networks, they try and replicate the brain, they try and replicate, well, on a much smaller scale, neurons and synapses, they do it in this nonlinear type of activation,

this really interesting structure and function combination that mimics that of neurons and synapses. And so we know that neurons and synapses and other types of cells lead to really cool stuff, you know, I'm sitting here and talking to you because my neurons and my glial cells and my interneurons and all that kind of cool stuff works together to allow me to speak, to think, to breathe, to do all the cool things that I can. And we're not at the stage where neuromorphic nanowires or other neuromorphic systems can do that. They can do a lot of really interesting stuff, they can definitely reduce the amount of energy requirements that we have, but we're definitely not at a stage where we can implement that. So why not try do it in neurons? We know that neurons can do it, let's try and do that. And that's what cortical labs is trying to do. So pretty much what we do very, very briefly is we take induced pluripotent stem cells. We take a blood sample from a donor, a human donor, very, very non-invasive, simple procedure. We take some blood like you would at a blood test or, you know, if you're donating blood. And the amazing biologists in our team managed to convert it into something called induced pluripotent stem cells, which means cells that can be reprogrammed to be any type of cell. And we do this, we change these iPSCs in induced pluripotent stem cells into something called neuronal cultures or neural cells that are very much cells that are in the cortex, very similar to cells that you and me have. In fact, if the blood sample is from you, those cells would be exactly, would have the exact genome that you have. So we've seen that in patients who have epilepsy, if we take their blood, the cells that we take also have epilepsy, which is really interesting. We can also take it from mice, embryos, or other types of animals, which a lot of other people do. And then what really is cool about in differentiated cortical labs is we teach them to play games. And we do that using the free energy principle, which, you know, where I had to bring in some active inference related stuff. And so that's sort of what we did in our first paper. We presented, we modeled this kind of pong environment, the game pong. Hopefully most of you know this game. And the neurons, we tell the neurons, we put them on this chip, similar to what I showed in the nanowires. And we tell them, you are the paddle, control the paddle based on this kind of structure, this setup that we give you, and try and predict where the ball is going to go. And the way we do that is we give them either an unpredictable feedback or a predictable feedback. And when they miss the ball, we give them this unpredictable random kind of feedback. And when they hit the ball, they get a predictable sensory feedback. And this fits quite nicely with the free energy principle, the idea that we're trying to minimize the randomness and try and get some predictable, interesting behavior in order to sort of maximize our reward and maximize our understanding of the environment. We also give a silent feedback where nothing happens. So the ball just continues just as a control. And what's really interesting, and I didn't actually include the graph, but I can refer you all to the papers. And I'm sure Brett's the CEO of the CSO of my company has spoken a couple of times here. So you may be familiar with that. But we see that when we give this predictable sensory feedback, the neurons learn to play this game very, very quickly, within five to 20 minutes of playing the game. And they play it at statistically significant, statistically better than average, more than chance would expect. And so this kind of encoding scheme where we're just giving this predictable versus unpredictable sensory feedback,

that's the only encoding that we're giving them, we see quite interesting support for the free energy principle being used in this kind of neural based system. So there's a lot of stuff that I've covered today. There's still so many questions left. I'm more than happy to answer those questions, ask, raise more questions. If anyone wants to ask me in person, this is my email here, add me on LinkedIn. I love to chat to anyone and everyone about this, you know, 40 minutes it took in the end, is not enough to cover everything that I did. But I hope it gave a nice overview and a nice start for people to go in and try and understand and ask questions. So thank you very much to my supervisor for my PhD, my current bosses and workmates and for you guys for having me and for listening. And I look forward to your questions. Thank you so much. That was absolutely fascinating. That was amazing. Thank you. I have so many different questions. I don't even know where to start. I mean, the main thing I was wondering about is when you were showing the networks with the MBAC test, which was super interesting. Yes. Nanowire networks. And you were talking about reward and punishment. I was quite interested in how to implement that. But then you showed the work at Cortical Labs with Pong and it's about unpredictable. So is it a similar type of reward and punishment that you did with your simulation, with your nanowire networks to that? So unfortunately, I wasn't aware of active inference for energy principle, all that cool stuff when I was doing my PhD. That was I was introduced to that at Cortical Labs. I would have loved to try that on these nanowire. I imagine it would be a fantastic way of doing this. We did something more similar to what you would imagine in gradient descent, but also this kind of feedback. So in a sense, we're trying to give a more not maybe not predictable, but we're trying to match what the output of the network is with what the expected output is. So maybe that is in a little bit like a predictable kind of trying to increase the predictability. But pretty much what we did is we changed the... Can you see my mouse or do I need the... Yeah.

Oh, you can't see mine. Okay, great. Which pretty much what we did was we changed the voltage of the desired output electrode. And so if the correct paths were being formed, because this is a physical system and it's a metal system at the end of the day, you get this kind of path of least resistance forming from where your sources, where your inputs are to where your drain is. The drain is pretty much just a grounding, you know, a zero voltage kind of thing. But you can play around with that voltage. It doesn't have to be zero. You can increase it, you can decrease it, you can increase and decrease the other ones. And so we implemented something similar to gradient descent on these voltages, on these electrodes. And when voltage is increased, for example, then you imagine there's less current or less current will flow to that electrode. And when voltage is decreased, more current will flow to the electrode. So on a very basic level, that's sort of what we implemented. But we saw that wasn't enough. That really wasn't enough to train it. It needed this rewarding kind of thing. So we needed to give some kind of threshold that when that's reached, we increase it. It was a bit more complex than that. But that's pretty much the idea behind what we did. Thank you. Okay, very interesting. Cool. Yeah. Does anyone else have any questions? Yeah, I would have one question. Really interesting. Please. Yeah. Thank you. Thank you.

Um, so you started your talk with motivating it with the energy consumption of the brain.

Correct. Really low. So can you give us a kind of rough estimate of how much these nano bio systems?

Right. They're tiny, tiny, tiny on the on the scales of watts, you know, we're talking even,

even significantly lower than that. Um, the only energy that's kind of being used in the system is

what you pump in from the, you know, the inputs that you give it. So you have to give some kind of

voltage as an input or, you know, and then some current is formed from that. But the system itself,

the beauty of this is because of it's a, you know, it's a fully connected kind of physical system,

we can solve the weights automatically using Kirchhoff's laws of current and voltage. It's

just a physics problem at the end of the day. And so physics just does it for us. This spreading out of

the voltage, the spreading out of the conduct of all that stuff, as I showed, if I go back here,

right, all this stuff that we see here, the spreading out of this kind of activation happens

inherently based on the structure of the network. And that's why it's so important to try and

understand that structure and combine it with a function. So when you think about it that way,

that's significantly lower energy requirements than trying to change each weight individually of each

one of these, you know, junctions over and over and over and over again, to better, you know,

get a better result. Okay, how many signups are typically could you realize in such a physical

system? For sure. In our simulations, we use significantly smaller simulations, because that's

what requires a lot of energy to simulate. We use about 100 to 700 wires in the actual system,

tens of thousands, if not hundreds of thousands of wires per system. The really other interesting

thing is, each one of these systems will be unique. So you'll have a slightly different structure,

slightly different kind of way that it's implemented. And a lot of people think, oh,

that would make it function significantly differently. But we don't see that we see,

you know, standard kind of characteristics, obviously, there's variance, but you know, the same is with

humans and with brains, you see very different structures and very different, you know, experiences

that humans have. And there is a variance in who we are. But for much of the case, our activity and

the way we function is quite similar at the end of the day. Okay. But in the systems, you actually really

realized physically, how many, so I guess if you have like 10,000 of wires, only a subset of those would actually form

correct functional potential synoptic connections, right? So there's probably a lot of

are two are like electrically isolated from each other, right? So I guess that's 100%. So

the really nice thing about these for now is that they're two dimensional. And so because they're

two dimensional, a lot more of the wires cross over each other in kind of separated regions. So you

wouldn't you wouldn't get, for example, the right region connected to the left region, right? And so

that makes it a lot simpler to model. When you start getting three dimensional models, you get this kind of,

you know, wrapping around and a lot of different connections connecting to each other. So there could be a

lot more connections or a lot less connections. It doesn't necessarily mean that when a wire

crosses over another wire, a perfect connection is formed. So it's quite difficult to expect this and

model it. That's why we have to rely on electrical characterization and seeing these kind of conductance

curves here, right? Comparing the conductance curves versus with the models that we have with the actual uh, the actual physical systems. That's how we do that kind of model.

Okay. Yeah. Thank you. And for the, for the dish brain, for the like biological cells.

So, I mean, the brain is actually fairly structured, right? In the end.

Yes, correct.

So are you working in the direction of also using a bit more of the brain's morphology?

Absolutely. And cell types and also layers. So we have a whole bunch of different cell types that we have. We do all sorts of combinations of excitatory inhibitory cells, glial cells, astrocytes, all that kind of stuff. Um, we're currently still also working on two dimensional layouts or two and a half dimensional layouts. But there's, we've had some collaborations with people, for example, John Hopkins University in America, um, which I believe I can talk about, um, hopefully I can talk about, uh, where we work with organoids, um, which are three dimensional, tiny little kind of cell structures.

They're not fully, you know, structured like what you'd see in any kind of brain organ structure, but they're a lot more similar to a tiny kind of brain structure. And so what we're really, I mean, my end goal in this is to try and stack chips together, you know, so a chip might be, have a two-dimensional structure, but then you can have it, um, touching another chip forming, you know, forming long connections across that kind of barrier with, with another chip. And then your first chip might act as some kind of sensory region and your second chip might be a processing region, right? And then you can form these pseudo structures in that way. Um, what's beautiful about neurons compared to nano wires, which I think is another reason, and I didn't touch upon this, is neurons self-organize after they're, um, placed down. So with nano wires, you put them down, they self-organize and they never move again. You know, they will never move again because they're silver wires and you cannot move.

Neurons automatically are constantly looking for connections, constantly changing their structure based on the input that is given to them. So, you know, they will form inherently structures as well.

Neurons automatically. Um, and we're really, we're exploring that. We're exploring how that happens when you're giving them predictable feedback, how that happens when you're giving them unpredictable feedback, how it happens when you give them no feedback at all. And these are all the kind of questions that we're trying to, you know, um, answer at Cortical Labs, which is, I think I find extremely interesting.

Mm-hmm. Okay. Thank you.

Neal Shubhler- It's interesting. Um, I'm really interested in the applications, which you also briefly touched on, um, in terms of when you said you get a donor with epilepsy and then you, um, their cells also exhibit epilepsy, which then leads me on to thinking about, well, the applications of this, could you potentially see this as sort of like a neural aesthetic or something where, um, say with someone with epilepsy, they have that little part of their brain with the high concentration of epileptic cells removed sometimes. I think that already happens now with epilepsy surgery. Could you see this almost of them being an implant where you've taken their cells, you've edited out the epileptic gene. If you, if it was able to be identified, then you

reprogram them into neurons basically without epilepsy and then sort of put them back into the bit that you could take out.

I think that would be absolutely fascinating. Um, I think there's so many different ways this can go down. Luckily, I'm not the one who has to decide that.

I have a lot of ideas myself, uh, but, uh, I think one really key thing that we, I should highlight is, you know, what you, what you mentioned is really, really interesting. And some people might think, oh, that's currently already being done in brain computer interfaces, right? There's a key difference between what we're doing and what brain computer interfacing is. Pretty much the idea with brain computer interfacing is you have a chip that reads the activity of the brain, right? And it spits out whatever that activity is. And then you have some very clever algorithms to understand what that original signal was in the brain. So for example, you might think of a tennis game and then you get some kind of electrical signal that represents a tennis game. And eventually these, uh, algorithms will pick up that you're thinking of a tennis game. Now that's really cool, really important, going to help a lot of people, but there's a significant limitation in that. Whereas we cannot, we don't know what the brain is doing still to create those images, right? Of a tennis game. Why are certain regions lighting up to represent that? And, and brain computer interfacing is not giving us any information about that. You know, it's just trying to reconstruct what is already being, taking place. And what we're trying to do is we're actually trying to figure out the algorithms and the encoding schemes, you know, whether it's free energy principle, whatever else might, might be, hopefully it is something as, as straightforward as the free energy principle, um, where we want to actually write those encoding processes into these neurons, teach them the output that we want them to give us rather than just interpret what the output is. So we're doing the encoding into the networks and the decoding, whereas the brain computer interface is sort of just the decoding. And so, you know, in an epilepsy kind of, um, situation, you might be able to have a chip like that in, in the brain. This is just, you know, pure conjecture, but it could be interesting. Um, and you feed in some kind of input into that chip, which programs the cells that it's, you know, that are being controlled by that to give a different output than the epileptic output that we're already seeing. Right. And so rather than just saying, oh, that's an epileptic fit, we can predict that it's like, okay, yeah, we can predict it, but now we can actually do something to change it. Right. We can actually change it with some kind of input. I like to imagine in a very futuristic kind of application of this, hopefully not too futuristic, but in a very futuristic, uh, application of this, you know, in the matrix when Neo is plugged in and he suddenly goes, you know, he, his eyes go blank and they're uploading all this stuff to him and he wakes up and he goes, I know Kung Fu. That's sort of the, the kind of very futuristic, uh, sci-fi application of, of what we're actually doing. We're actually encoding Kung Fu into these cells and then trying to replicate that. Yeah. That is absolutely fascinating. Yeah. I was actually going to ask you sort of, what are you most excited about in this field? Like what do you see coming in the future for you? Would it be sort of the brain computer interface space? I think the brain computer interface space is fascinating. You know, we're seeing really interesting companies, even local Australian companies like Synchron overtaking

these multi-billion dollar, uh, organizations in, in this space. Um, we're going to see a lot more interesting kind of, uh, predictions of what people who might not be able to communicate normally, um, and, and reconstructing their thoughts and, and what they're actually trying to, you know, represent. Um, and so that's going to transform a lot of people's lives. It's going to transform medicine in that sense. But what I'm much more interested in, you know, it's still very interesting and it's going to be really cool in the near future, but what I'm much more interesting in, in is this idea of, can we write things into neurons? Can we program neurons to give us the kind of outputs that we want rather than just, you know, interpreting what they're already doing. And that's what we're doing at cortical labs in the long run. Um, you know, it might take a long time to get there, but really, really interested to do that. Very cool. And then something which obviously people are talking about a lot is like the ethics of AI and the explainability and the transparency of AI, which is a current problem with the, you know, black box methods that we have. One of the things I found interesting about active inference is people talk about how you can actually have much more transparent and explainable AI potentially. And I see what you're doing with the, do you think with the nanowire networks, you also have the opportunity for that because you can actually go in and don't know, we didn't get onto how you actually visualize and analyze them, which is interesting, but like, do you think that this type of AI would be more explainable than sort of traditional artificial neural networks? Oh, without a question. I mean, you know, the reservoir computing approach that I, that I touched on does use that black box approach, which is why I didn't really like it. I used it for the first couple of papers that I worked on and then was sort of like, this is not how the brain works. Maybe it is, maybe there are some aspects of it, but the brain is not a black box. I mean, to us it is, but you know, there are so many little pieces that work together to, you know, create these kinds of emergent properties and there's rules that govern that and there's structures that govern that. And so there must be a way to implement those kinds of rules into something, into these systems and then be able to actually see them learning. And that's, you know, that video that I showed, you can see how it's changing over time, which is very, very different to these black box approaches where we don't even know why weight, you know, why different weights changing results in this kind of learning. I think in neurons, you can do all sorts of cool calcium imaging. You can do infrared kind of microscopy. I'm not a biologist, so I'm sorry to my team if I was wrong about that, but you know, we have them on these transparent cells, arrays, so we can actually see, we can take images, you know, we can put them in the incubator, play a bunch of long games, take them out, take some images, see how they've shifted, see what's happened and try and correlate that with the activity. That's a much more transparent way of learning, in my opinion. Very cool. So it's almost like we have more energy efficiency, we have more explainability, and we have like these opportunities for personalization, almost like with your own cells in the chip, which I think is really cool. It's a very exciting direction, I think, for lots of different technologies, I guess, not just brain-computer interfaces as well. For sure. And, you know, I think what's really interesting is how we can combine

these technologies like neuromorphics with neurons. Neuromorphics are trying to replicate neurons. Is there a way we can use neuromorphics to kind of act as like, you know, eyes, you know? Our eyes do this pre-processing kind of activity where you take in a light signal, an electrical signal, and convert it through some kind of cell, you know, activity into a more understandable signal for your occipital lobe and your other regions of your brain? Can we use neuromorphic systems like nanowire networks to take in a raw input and output a more biological light signal, which then neurons would be a lot more receptive to? You know, I said we'd really like to encode these, write things into the neurons and get them to do things that way. But the question of how to do that is a really difficult question. Otherwise, people would have figured it out by now. We have a lot of theories, but one of those might be as simple as let's use these technologies that are taking in simple or complex inputs and making them into a more biological-like output based on just their inherent properties and then feed that, you know, interface with a biological system. Who knows? Maybe that'll be a really cool way of getting that kind of encoding and writing in method. So that's something that I'm really interested in, in marrying those two worlds together as well. Okay. I'm conscious I don't want to go too much over time, but I do have a little bit more time. Sure.

I've got a little bit more time. I've actually got as much time as you need to.

That's very cool. Anyone else please jump in with questions, but I did just want to touch on something which I'd heard about with your work before, where you mentioned about the quantum aspects of the silver nanowire network. Yes.

Not a quantum computer scientist by any means, but I'm interested in this because there's sort of this debate in neuroscience about whether or not there's a quantum brain, like the brain, quantum. And maybe, you know, for olfaction, there's been some suggestions, some theories about that's how we smell with quantum tunneling and the olfaction receptors, but the rest of the brain, people seem to be, the jury seems to be still out. And I'm just interested in your network seem to use, you mentioned that there was like some quantum tunneling forming the bridges potentially.

Is that your work or was it just, you know, something which is interesting, but wasn't really you examined? So it was actually more of my colleagues, Joel, and we have a paper in Nature Coms, Nature Communications about that. So I can, I can definitely share that with people. If you Google my name, it's the first thing that will come up. But yeah, I love that we're covering all the buzzwords here. It's fantastic. So yes, the quantum tunneling, or we call it electrode tunneling. We see this behavior. It's sort of the only way we can explain this. And so that's sort of why we've hypothesized and Joel's shown some evidence for this. But the idea is that you start with a no bridge here on the left, for example, and over time you get this fully connected bridge, but that doesn't, it's not like an immediate kind of response that takes time to sort of the ions flow slowly through that. It's quite quick, but it's not an immediate response. But what we start to see is if, for example, the bridge is kind of semi-formed and let's imagine these two dots here aren't there, these two little balls at the end or ions aren't there, you can see, you start seeing these weird fluctuations of

conductance before that switch is fully formed. And that's explained by this quantum tunneling or electro tunneling of electrons jumping through the gap between the, whatever leftover gap is, is there just by chance. And that actually accounts for really interesting behaviors that I haven't covered today. I don't really fully understand them myself. That's not my area of expertise, but Joel has explored them and has some really interesting theories about how that might affect the networks. One example is this idea of criticality. So criticality is sort of the theory that, you know, biological systems or a lot of systems, actually real world systems, have maximal information processing at the edge of criticality or the edge of chaos. So a chaotic system is one that is sort of completely unpredictable, has full chaotic activity. And an audit, on the other hand, an audit system is something that nothing interesting can happen because you can, you know exactly what's going to happen. It's fully predictable, you know, every single aspect of it. But somewhere in the middle, somewhere in between where you're not too chaotic, there's still quite a lot of unpredictable behavior, but there is a predictability to that chaoticness. It's called the edge of chaos. That's where things seem to thrive, life seems to thrive. And we've seen criticality in nano wire networks. We've seen criticality in our neurons as well in cortical labs. And one of these ideas is that potentially the electro tunneling might help tune the system into this critical state.

We don't really fully understand it. If you're really interested in that, the paper is a really cool summary of that, I guess.

I'll get the link off you and I'll make sure you put it in.

Sounds great.

YouTube description. Yeah, because I'm really interested in sort of what these synthetic systems can then tell us in turn about the brain for those sort of things we've been observing about for a long time, like whether or not the brain uses quantum. And it's interesting actually as well, because there was a Loihi chip and they used that, the neuromorphic nose, and that performed quite well. So it would be interesting if you could sort of combine with these types of networks that maybe do have the quantum effects and sort of test out these more quantum theories of olfaction. That would be pretty cool.

For sure. I think there's so much to explore here. Look, I have maybe been unreasonable to some of the neuromorphic chips like Loihi in this talk. I know I only briefly spoke about it. There's definitely space for that. They're doing amazing things. There's some really cool learning algorithms and some really awesome people working on this stuff. I think my perspective from this is that we want to try and replicate more brain-like learning and more brain-like activity. And for that, we need a structure that is more brain-like as well. The structure, function, behavior, and relationship is key, in my opinion, to unlocking the emergence, the emergent properties in the brain.

And so a lot of those chips lack that. And hopefully people will work more and more on these brain-inspired systems

and start to consider the structural element of it as well, because it is extremely important.

And I guess you've spoken about the nanowire networks and you've also spoken about the more neuronal

networks.

Where's your work going in the direction? Are you moving away from the nanowire networks? Or can you see a hybrid

of both actually being quite useful in the future?

Yeah, so I think I touched on that briefly, you know, marrying the two nanowires and neurons together as sort of this like encoding bridge or something like that. That's, you know, unofficially something that I'm interested in. I'd love to explore that, whether or not that's something that I'll have the time and the resources to do. We'll see. But that's, you know, definitely where my interests lie.

I'm very happy to be able to apply, you know, my knowledge and my expertise in these kind of algorithm development and brain learning in actual neurons. So that's sort of what my focus is on at the moment.

But I'm still very much in touch with my supervisor and my team at University of Sydney. And, you know, we've got a paper coming out later this year or I guess next year. Or did it just come out now?

I think it just recently came out actually. What am I talking about? Gosh, what year is it?

It just recently came out in Nature as well. So Nature comes. So yeah, go check that one out.

Very cool. Okay. Has anybody else got any questions before we wrap up?

Cool. Okay. I've got one last question. One thing I'm always curious about is if there's any sort of major problems in your field that you want someone to go out and solve almost like, because anyone might watch this and they might be working on some obscure little problem. And do you have any asks?

So it could be, you know, this mathematical theorem or, you know, create some hardware that we can do this weird thing with. Like what would be your main asks, I think, for anybody listening who might work on something weird and wonderful? So I think for that, let me just go back to the final slide again.

I know I'm jumping around back and forth. But I think what I'm working on most at the moment and what I think is the most important thing for our work and for anyone who's working in this space is figuring out that encoding paradigm. What is it that the eyes do, that the ears do, that the nose does? Actually, I think the nose does it least, but that our sensory organs do so well to encode, you know, what might be seen as nonsense information, you know, there's obviously structures to it and it's not nonsense, but it's, you know, light or sound or very different forms of information into something that the brain can understand and make inferences about, you know, and not just make inferences, build models of the world to try and replicate it. You know, active inference is all about trying to understand that. How do we do that? Obviously, that is a huge question. But in a simpler way, how do we encode a simple electrical impulse or a string, a temporal sequence into something that neurons can understand? Okay. And we can actually elicit different outputs from them.

Very cool. Okay. So if anybody's listening and they're working, he is getting in touch with Alon because that would definitely catalyze the field, I guess, solving those kinds of questions and problems. If we can solve that, then yeah, I mean, we'll be having very different conversations in the next few years. I think very, very exciting conversations. Yeah. Well, hopefully the more stream will still be going. We can bring you back. I would love that. I would absolutely love that.

Okay. Right. I guess we'll end there. Have you got any last comments or anything before we end the

recording? Just feel free to email me, reach out. I think science needs to really be open and collaborative. And, you know, I do work for a company, but I'm also still very much about open source work. And, you know, I will and add me on LinkedIn and let's just keep chatting. Let's just keep answering these questions. Great. Thank you so much. And thank you everybody for listening and for joining the call today. And thank you for having me. See you later.